



Robust Foundation Models for Causal Inference

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Motivation

Estimating the conditional average treatment effect (CATE) (defined as $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$) is critical for personalized decision-making, yet challenged by unobservable counterfactuals, small sample sizes, and noisy covariates. Unlike traditional machine learning methods that require gradient optimization and lack causal inductive biases, causally-pretrained tabular foundation models (e.g., CausalPFN, Do-PFN) embed structural priors directly into their weights. This enables zero-shot, in-context inference that is particularly suitable for small-sample regimes. Our study addresses three questions for tabular foundation models: (1) how a generally pretrained tabular foundation model compares with a causally-pretrained model for causal inference; (2) how the performance of a tabular foundation model scales with training data; and (3) how robust a tabular foundation model is under covariate contamination. Our results show that a generally pretrained tabular foundation model is competitive with causally-pretrained models for causal inference. At the same time, scaling training data leads to improvements in reducing worst-case risks, while increasing noise contamination in training data leads to performance degradation.

Methods

Prior-Data Fitted Networks (PFNs)

- Aim: PFNs are transformer-based tabular foundation models that carry out approximate Bayesian inference in a single forward pass without gradient updates [1, 2].
- Training: PFNs are trained on millions of synthetic datasets sampled from a predefined prior over $p(\theta)$ data-generating processes to predict held-out targets.
- Inference: The real-world dataset D is fed entirely to the PFN. The output directly approximates the posterior predictive distribution of test the target y , amortizing Bayesian model averaging via a single forward pass:

$$p(y \mid x, D) = \int p(y \mid x, \theta) p(\theta \mid D) d\theta$$

Models for Evaluation

We consider three PFN models for evaluation: 1) **TabPFN v3** is a generally pretrained tabular foundation model for in-context learning; 2) **CausalPFN** is a causally-pretrained tabular foundation model with state-of-the-art performance for causal inference; 3) **Do-PFN** is another causally-pretrained tabular foundation model specializing in inference based on structural causal mechanisms. For other details of the three PFN models, please see Table 1.

Table 1: The three tabular foundation models

Model	Pretraining Prior	CATE Estimation Mechanism
TabPFN v3 [3]	General tabular prediction	Via T-learner: $\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x)$
CausalPFN [4]	Synthetic causal DGPs	Direct estimation
Do-PFN [5]	Synthetic do-SCMs	Predicts $p(y \mid \text{do}(t), x)$

Evaluation Metric, Reference Baselines, and Datasets

- Evaluation metric: We consider precision in root Precision in Estimation of Heterogeneous Effect (root-PEHE)

$$\sqrt{\text{PEHE}} = \sqrt{\frac{1}{n} \sum (\tau_i - \hat{\tau}_i)^2}$$

- Reference baselines (Independent of method or Training Size):
 - Predict-0 (dashed line, $\sqrt{\mathbb{E}[\tau^2]}$): It is the line for the model that always predicts $\hat{\tau} = 0$.
 - Oracle ATE (dotted line, $\text{sd}(\tau)$): It is the line for the model that delivers Average Treatment Effect.

Benchmark datasets

We consider the four datasets (shown in Table 2). These four datasets have various sizes of observations and numbers of covariates, and are frequently used for testing methods in causal inference.

Table 2: Benchmark datasets.

Dataset	Train Pool	Test Set	Covariates (X)	Treated %
IHDP	672	75	25	18.30 %
ACIC 2016	4,321	481	58	17.80 %
Lalonde CPS	14,559	1,618	8	1.06 %
Lalonde PSID	2,407	268	8	5.86 %

Discussion

- The generally pretrained tabular foundation model, TabPFN v3, is competitive with two causally-pretrained models, CausalPFN and Do-PFN, across the four benchmark datasets.
- Across all three models and four datasets, scaling training data leads to initial gains before hitting dataset-specific saturation regions; nevertheless, further increasing in training data size can still improve prediction variance and reduce worst-case risks.
- While all three models fail under extreme noise, this degradation is different across the four datasets. For example, in ACIC 2016, severe contamination can lead models to a failure that is worse than a baseline assuming zero treatment effect.
- The two causally-pretrained models perform differently. CausalPFN performs similarly to TabPFN v3, while Do-PFN does not perform well across the four benchmark datasets under the conditions we set for the experiments.

Reference List

- [1] N. Hollmann, S. Müller, K. Eggenberger, and F. Hutter, "TabPFN: A transformer that solves small tabular classification problems in a second," in *Proceedings of the 11th International Conference on Learning Representations (ICLR)* (2023).
- [2] N. Hollmann, S. Müller, L. Purucker, A. Krishnakumar, M. Körfer, S. B. Hoo, R. T. Schirmer, and F. Hutter, "Accurate predictions on small data with a tabular foundation model," *Nature* 637, 319 (2025).
- [3] L. Grinsztajn et al. (Prior Labs Team), "TabPFN-3: Technical report," *arXiv:2605.13986*.
- [4] V. Balazadeh, H. Kamkari, V. Thomas, B. Li, J. Ma, J. C. Cresswell, and R. G. Krishnan, "CausalPFN: Amortized causal effect estimation via in-context learning," in *Advances in Neural Information Processing Systems 39 (NeurIPS)* (2025).
- [5] J. Robertson, A. Reuter, S. Guo, N. Hollmann, F. Hutter, and B. Schölkopf, "Do-PFN: In-context learning for causal effect estimation," in *Advances in Neural Information Processing Systems 39 (NeurIPS)* (2025).

Experiment

Experiment 1: Baseline CATE Estimation

- Purpose: Establish how the three tabular foundation models perform on CATE estimation with each dataset.
- Settings: Use the full training data for each dataset alongside fixed test sets.

Table 3: Oracle-normalized root-PEHE across full training pools

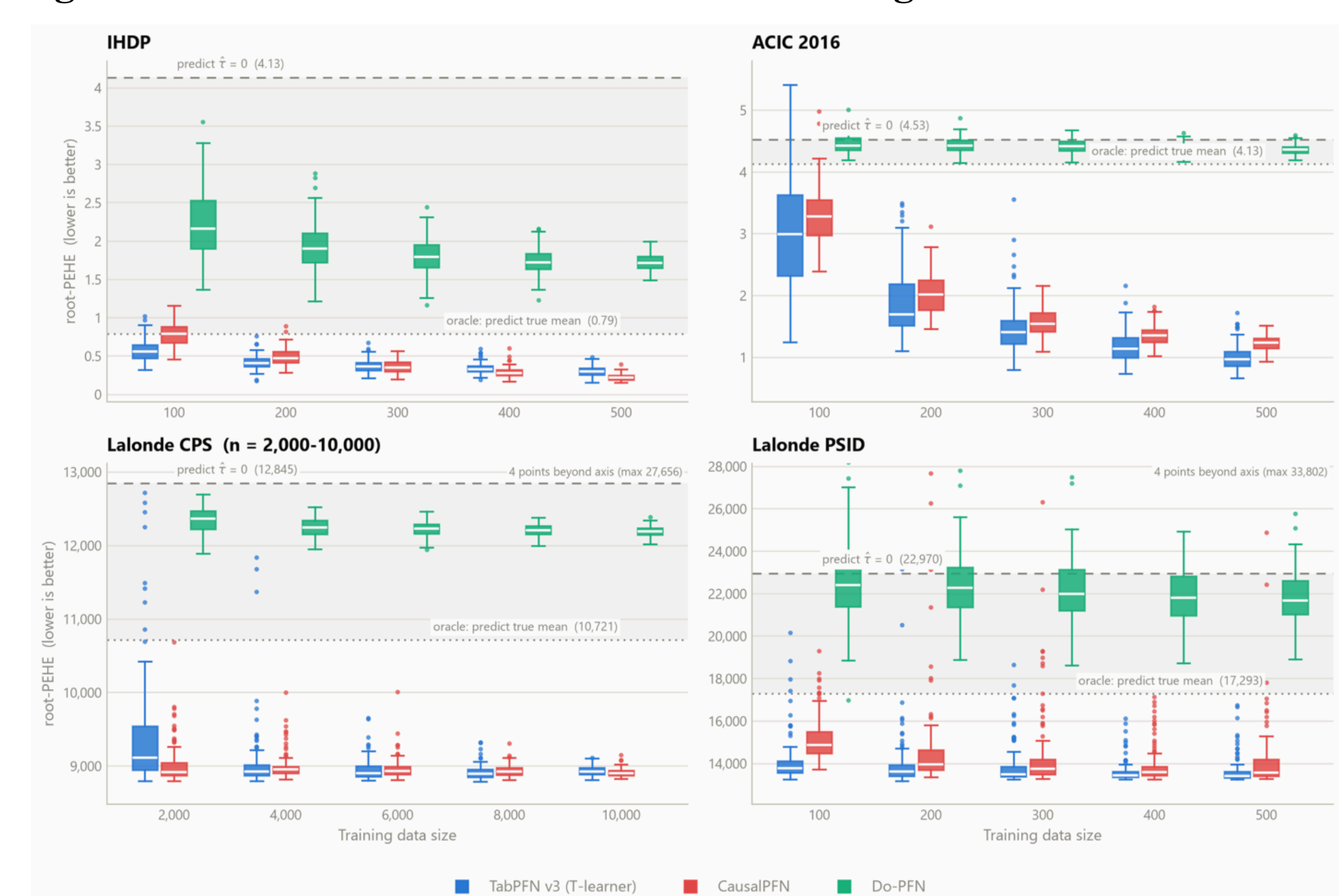
Model	IHDP	ACIC 2016	Lalonde CPS	Lalonde PSID
TabPFN v3	0.417	0.102	0.837	0.770
CausalPFN	0.197	0.176	0.827	0.784
Do-PFN	2.125	1.040	1.134	1.225

- Results: From Table 3 we can see TabPFN and CausalPFN perform similarly while Do-PFN consistently fails to obtain true heterogeneity.

Experiment 2: CATE vs. Training Size (Sample Efficiency)

- Purpose: Evaluate how performance scales with training size and determine where accuracy gains saturate.
- Settings: Models are evaluated across sample sizes ranging from 100 to 500. Lalonde CPS is scaled from 2000 to 10000 due to sparsity. The experiment is repeated 100 times.

Figure 1: CATE estimation error versus training-set size.



- Results: From Figure 1 we can see the median root-PEHE values for the three models drop sharply before hitting dataset-specific saturation ceilings. Beyond this saturation point, increasing training data improves prediction variance and reduces worst-case risk.

Experiment 3: CATE vs. Covariate Contamination (Robustness)

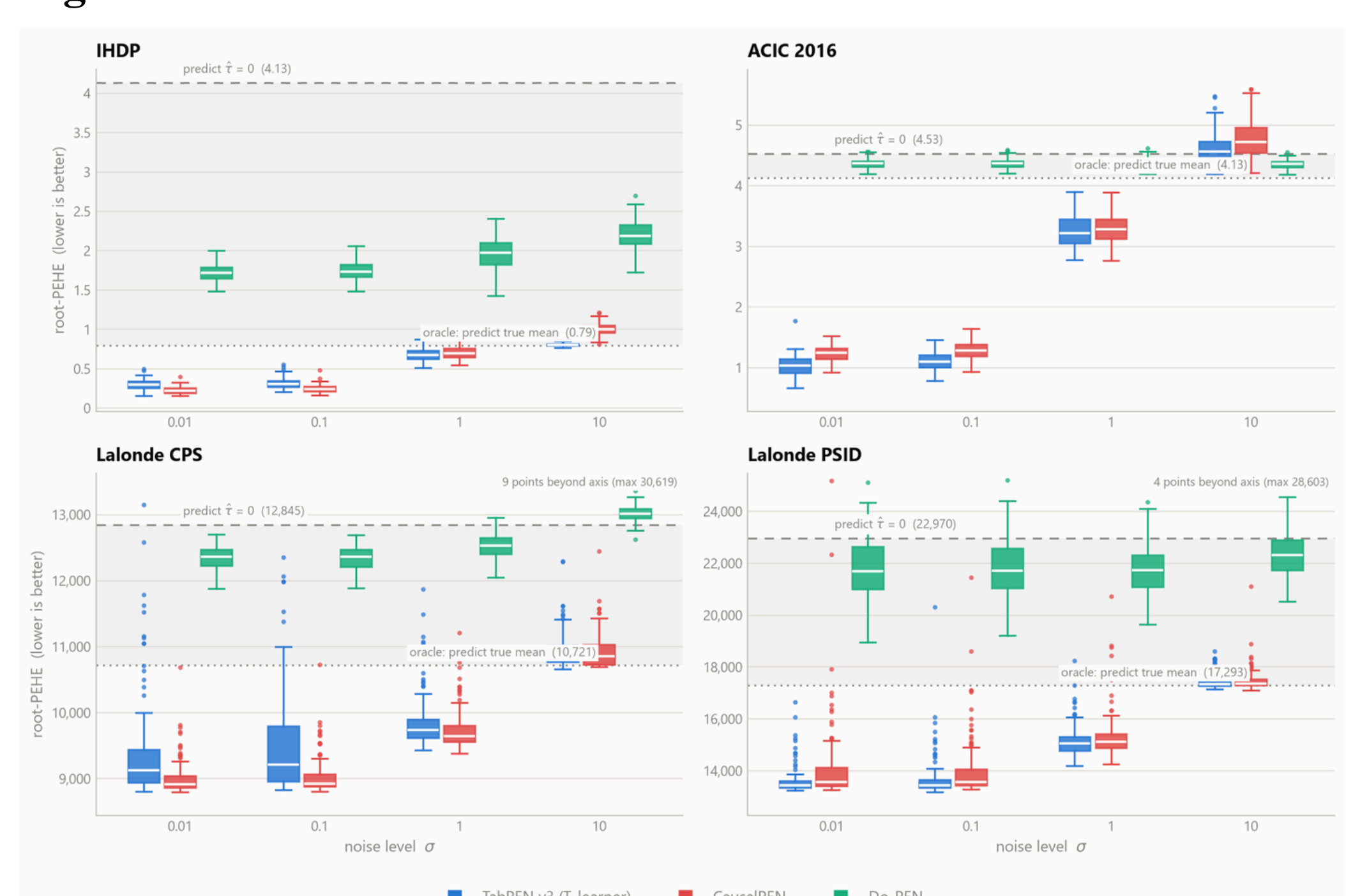
- Purpose: Test model reliability when observed covariates are subject to structured measurement error.
- Settings: Fixed sample sizes (n=2000 for CPS, 500 for others). Noise $\sigma \in \{0.01, 0.1, 1, 10\}$ is added strictly into covariates:
- Numeric injection:

$$x \leftarrow x + \sigma \cdot \text{sd}(x) \cdot \epsilon \rightarrow \text{Reliability} \frac{1}{\sqrt{1 + \sigma^2}}$$

- Categorical injection: Resampled from empirical marginals with probability

$$p = 1 - \frac{1}{\sqrt{1 + \sigma^2}}$$

Figure 2: CATE estimation error versus covariate contamination.



- Results: As shown in Figure 2, causal priors provide no robustness advantage, as TabPFN and CausalPFN degrade at nearly the same rates. Under extreme noise, all three models tend to produce estimates similar to the Oracle ATE baseline.